Design of small-scale quadrotor unmanned air vehicles using genetic algorithms

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Abstract: In the current paper, a new design methodology was proposed to automate the component selection, layout design and geometric sizing of small-scale quadrotor unmanned air vehicles (UAVs). Instead of using the swash plate mechanism employed in most rotary-wing flight vehicle, the UAV was controlled by four fixed-pitch rotors/propellers. The objective of this design optimization problem is to select the most suitable set of components from a range of commercial-off-the-shelf products, and organize the vehicle components/payloads, such that the resulting flight vehicle has the most compact overall size without violating any given physical or mission constraint. Genetic algorithms are employed as the optimization engine because of their proven results.

Keywords: unmanned air vehicle, design optimization, genetic algorithms

1 INTRODUCTION

In recent years, there has been increasing interest in the research development of small-scale unmanned air vehicles (UAVs) and micro-air vehicles (MAVs) (Morris [**1**] Rais-Rohani and Hicks [**2**]; Grasmeyer and Keennon [**3**]; Frulla [**4**]; Ng and Leng [**5**]; Watkins [**6**]). MAVs belong to a class of flight vehicles that are very much smaller than conventional aircraft. Though there are no exact size and weight definitions, the typical maximum linear dimension of existing MAVs is in the range of 0.50 m and below. The definition employed in the research program of US Defense Advanced Research Projects Agency (DARPA) limits this aircraft to a size less than 15 cm in length, width and height.

As the war against terrorism intensifies, military forces are turning to technology for cutting edge equipment such as UAVs and MAVs so that they are constantly ahead of their adversaries. These unmanned flight vehicles can provide military surveillance teams with critical intelligence that could save valuable human lives from a possible terrorist attack.

UAVs are greatly sought after in non-military applications as well. At times of industrial accidents or life-threatening situations, they can provide rescue teams with critical close-up information that will help the rescue teams perform better in their operations. For example, UAVs equipped with infrared cameras are able to locate trapped victims in burning buildings. They are able to manoeuvre through buildings to find sources of dangerous toxins or chemicals at the sites of industrial accidents.

In the current work, the emphasis is on the use of artificial intelligence to automate the design of a small-scale quadrotor UAV/MAV. Rotary-wing UAV/MAV has many strong advantages, such as the ability to hover at a stationary spot, and manoeuvre in densely populated areas with buildings in close proximity.

A low-cost approach to the development of an UAV can be achieved by integrating the smallest commercial-off-the-shelf components (such as propulsion systems, sensors, power source, video transmitter, and video camera) available. However, faced with a wide range of components from different vendors, it is a daunting task to select the best combination of components, and position them in order to obtain the most compact vehicle, and still satisfy the physical and mission constraints. This work

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is an extension and improvement of the previous work by Ng and Leng [**7**], to address these problems. Genetic algorithms (GA) are adopted in the current work, as they have proven themselves to be a best and powerful search engine in many engineering design problems (Tan *et al.* [**8**]; Fonseca *et al.* [**9**]), including conventional-sized helicopter design (Crossley and Laananen [**10**]).

2 DESIGN SURVEY

Most common rotary-wing flight vehicle in operation is probably the Igor Sikorsky's single main and tail rotor configuration. The main rotor provides the lift, and the tail rotor counteracts the torque generated by the main rotor, and hence, prevents the vehicle from spinning around in the main rotor axis.

The control is performed via a swash plate and linkage mechanism. Using the collective control, the entire swash plate assembly is raised as a unit. This changes the pitch of both the blades simultaneously, and the aircraft will climb or descend accordingly. Cyclic control pushes one side of the swash plate assembly upward or downward. This changes the pitch of the blades unevenly depending on where they are in rotation. Thus, the rotor has a greater angle of attack on one side of the helicopter and a lesser angle of attack on the opposite side. This allows the aircraft to move forward/backward or sideways.

It is not easy to fabricate and assemble the swash plate mechanism, especially for small-scale rotary-wing flight vehicle whose rotor size is less than 1 m. The use of high precision machining process to fabricate such small-scale mechanism results in high production cost. Another problem with this kind of mechanism is that through prolong usage, mechanical wear results in looseness or breakage in the linkages, causing inconsistency and uncertainty in control.

Another type of rotary-wing vehicle uses the tandem rotor configuration. The tandem rotor configuration basically consists of two main rotors that rotate in opposite directions to cancel each other's torque reactions, and thus, there is no need for a tail rotor. One classic example of such configuration is the well-known military helicopter Boeing CH-47 Chinook. However, the swash plate and linkage control mechanism is still required to control the vehicle's motion. The coaxial-rotors configuration works similarly to the tandem rotor configuration, except here, one rotor is directly above another, sharing the common rotational axis.

In quadrotor configuration, there are four rotors arranged in a diamond-shape formation. As the two pairs of rotors are of oppositely rotating types, the vehicle is balanced without the use of a tail rotor. In order to eliminate the swash plate and linkage mechanism, fixed-pitch rotors or propellers were used to replace the variable-pitched rotors. The translational motion can be achieved completely by varying the thrust of a specific propeller without swash plate and linkages. Thus, this configuration is chosen in this study.

3 PROBLEM FORMULATION

3.1 Design problem formulation

In the optimization design problem, the propulsion, sensors, radio receiver, and other components are modelled as rectangular blocks. The dimensions, length (L), breadth (B) and height (H) of each block are defined by the smallest rectangular box that can enclose the component completely. These dimensions, along with the mass, centre of gravity, voltage requirement, and power consumption information, are input to the optimization computer program.

The location of each component's centre of gravity, $CG_{component}$, is defined with respect to a Cartesian coordinates system whose origin is set at the centre of the rotor formation (Fig. 1). There are three possibilities to mount a component: BL, BH or HL plane. The mounting plane design variable is defined as 1 for BL plane, 2 for BH plane and 3 for HL plane. For each mounting plane, the component can be orientated with respect to the X-axis in two different ways. For example, if the mounting plane chosen is the BL plane, the component can be oriented either with the BH plane or HL plane facing towards the positive X-axis. Therefore, there are altogether six possible ways to mount a component into the flight vehicle, defined by three mounting planes and two orientations design variables (Fig. 2).

Selection of propulsion set is a tricky job as the more powerful propulsion sets can generate more thrust, but they are usually bigger, heavier, and consume larger currents. Instead of having only one propulsion set choice variable for all the four propulsion sets needed, it would be better to have two separate propulsion choice variables, one for front/back pair and one for left/right pair. As seen in Fig. 3, this would allow a more compact configuration, if one of the two

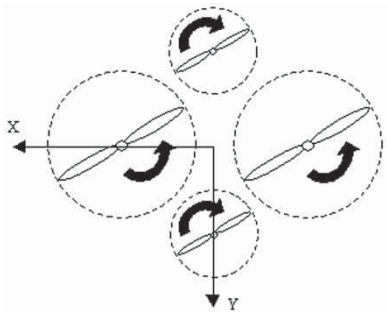


Fig. 1 Quadrotor layout configuration

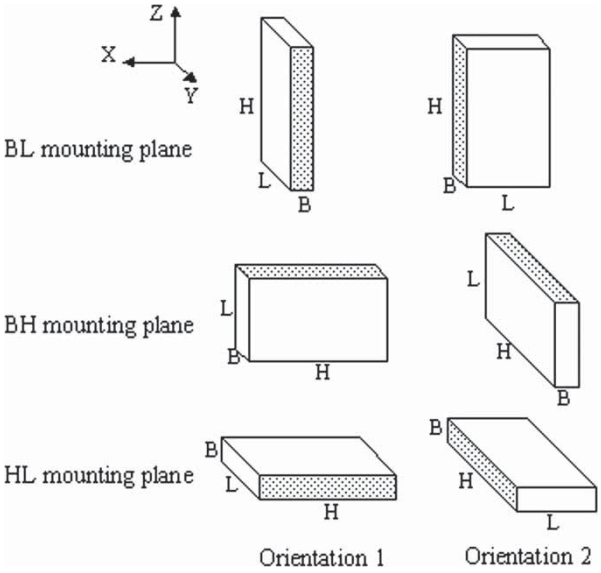


Fig. 2 Mounting plane and orientation of component definition

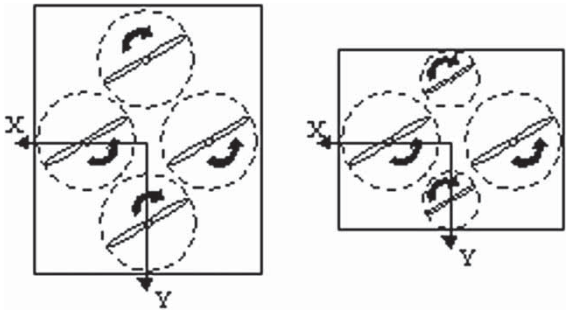


Fig. 3 Comparison of two possible quadrotor layout configurations

pairs has smaller propellers, and could still contribute sufficient lift to make the aircraft airborne.

Design variables used in the design problem are the component choice, mounting plane, orientation, and the (x, y, z) position of each component's centre of gravity, $\mathrm{CG}_{\mathrm{component}}$, in the design space. Another new design variable introduced here is the percentage usage of the maximum allowable current for each electric motor in the propulsion set, when the aircraft is hovering. Each motor has a maximum allowable current consumption limit to prevent it from having a premature burnout. The reason for this new variable is because the choice of propulsion set is a variable, and different set has different current consumption and thrust output. Moreover, it should not be assumed that each motor would be drawing its maximum allowable current to produce its maximum possible thrust in order for the vehicle to become airborne and stay hovering at a stationary spot. Another reason is because the electric power source, which is a battery pack has a finite current output that may be insufficient to meet all the electric motors' maximum consumption. Furthermore, if the two pairs are of different motor/propeller combinations, the more powerful pair must consume less current to balance the smaller torque created by the weaker pair.

Some of the necessary information about the propulsion set required by the design optimization are the minimum and maximum operating voltages for the motor, and the voltage versus current constant, K_{v}. In addition, the thrust versus current constant, K_{thrust} and torque versus current constant, K_{torque} of the propulsion set are also essential. The current, I, used by each motor is given by

$$I = \text{percentage current usage} \times \text{maximum allowable current} \tag{1}$$

where

$$\text{maximum allowable current} = \frac{\text{maximum allowable voltage}}{K_{\mathrm{v}}} \tag{2}$$

The thrust generated by the motor is

$$\text{Thrust} = I \times K_{\mathrm{thrust}} \tag{3}$$

and the torque produced is

$$\text{Torque} = I \times K_{\mathrm{torque}} \tag{4}$$

Propulsion set with higher K_{thrust} seems better as it produces more thrust with the same current consumption as another propulsion set of lower K_{thrust}. However, they usually have a larger propeller and a greater reaction torque.

It is desirable to minimize the overall dimensions of the vehicle's fuselage, X_{fuselage}, Y_{fuselage}, and Z_{fuselage}. This ensures that the frame that houses the components will be minimal, and the overall weight of the flight vehicle will be reduced. Thus, this objective will result in a compact UAV/MAV that is easier to store, lightweight, and more difficult to detect. In addition, it is also desirable to minimize the overall dimensions of the vehicle's propellers, $X_{\mathrm{propeller}}$, $Y_{\mathrm{propeller}}$, and $Z_{\mathrm{propeller}}$ (refer to Fig. 4). This ensures that the propeller chosen will produce enough thrust but not larger than necessary. Thus, the optimization problem is formulated as

$$\text{minimize } X_{\mathrm{fuselage}} \times Y_{\mathrm{fuselage}} \times Z_{\mathrm{fuselage}} + X_{\mathrm{propeller}} \times Y_{\mathrm{propeller}} \times Z_{\mathrm{propeller}} \tag{5}$$

subjected to the following constraints on the following:

(a) overlapped regions;
(b) maximum $Z_{\max}$ boundary;

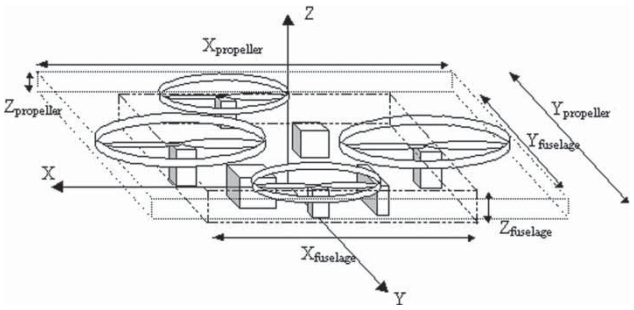


Fig. 4 Definitions of overall dimensions of rotary-wing UAV/MAV

(c) inter-propeller distance;
(d) overall CG location;
(e) minimum voltage and current of power source;
(f) minimum lift-to-weight ratio;
(g) minimum flight time;
(h) balanced yaw moments.

3.1.1 *Overlapped regions constraint*

It is apparent that no more than one component can occupy the same physical space. Thus, it is necessary to enforce this constraint in the design optimization. This constraint is incorporated into the objective function as a penalty function, whose penalty value is equal to the volume of overlapped regions between the components (Fig. 5). This ensures that as the components are positioned closer together, they do not actually cut into one another's region. The constraint function is given as

$$\text{Vol}_{\text{overlap},i} = 0 \tag{6}$$

where $\text{Vol}_{\text{overlap},i}$ is the amount of overlapped volume experienced by the ith component, $i = 1, \ldots, N$. The penalty function associated with this constraint is given as

$$h_1 = \sum_{i=1}^{N} V_{i,\text{overlapped}} \tag{7}$$

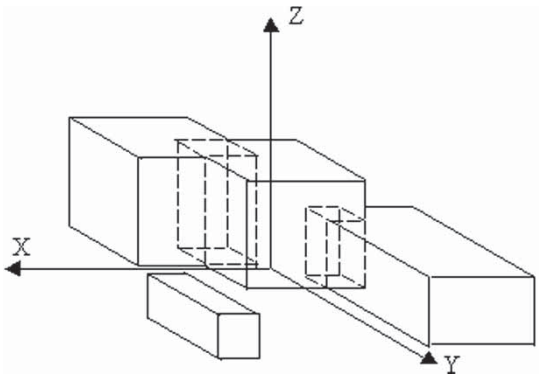


Fig. 5 Overlapped regions constraint

3.1.2 *Maximum $Z_{\max}$ boundary constraint*

Another geometric constraint that needs to be imposed in the optimization problem is regarding the plane where the rotor blade or propeller sweeps through, $Z_{\max}$. It is important that no component is placed in the design space such that it protrudes above this plane.

The penalty incurred under this violation is equal to the volume of component that protrudes above the main rotor/propeller plane, $Z_{\max}$. Thus, the constraint function is given as

$$\text{Vol}_{\text{protrude},i} = 0 \tag{8}$$

where $\text{Vol}_{\text{protrude},i}$ is the amount of protruded volume above the rotor/propeller by the ith component, $i = 2, \ldots, N$. If this constraint is violated, the penalty function is

$$h_2 = \sum_{i=1}^{N} V_{i,\text{protrude}} \tag{9}$$

3.1.3 *Inter-propeller distance constraint*

In the quadrotor configuration, it may not be necessary for all the four propulsion systems to be identical. In order for the flight vehicle to be balanced without any resultant pitching and rolling torque, only the front/back pair, and the side pair must consist of identical motors and propellers, and positioned in symmetry. Thus, each propulsion pair has only one location variable.

Though the overlapped region constraint enforces the propellers not to overlap one another, there is still a need to ensure that there is sufficient clearance between the propellers. Thus, this additional constraint is defined as

$$\delta_{\text{prop}} \geqslant \delta_{\text{prop,s}} \tag{10}$$

where δ_{prop} is the minimum clearance between propellers obtained by optimization and $\delta_{\text{prop,s}}$ is the stipulated clearance. If this constraint is violated, the penalty value is given by

$$h_3 = \delta_{\text{prop,s}} - \delta_{\text{prop}} \tag{11}$$

3.1.4 *Overall centre of gravity constraint*

It is possible to manually arrange the various components such that the final configuration is as compact as possible. However, the resulting overall centre of gravity of the flight vehicle may end up in an undesirable location that will render it difficult to control (Prouty [**11**]). The choice of the overall CG location depends on the type of control systems implemented or the designer's personal preference. It is highly inefficient,

if not impossible, to adopt a manual trial-and-error arrangement of the components to obtain the most compact possible UAV that will also satisfy the overall CG location constraint.

Once the mounting plane, orientation and position of the components are set, the overall centre of gravity of the rotary-wing UAV can be calculated as follows

$$\mathrm{CG}_{i,j} = \frac{\sum_{j=1}^{N}(w_j \times d_{i,j})}{\sum_{j=1}^{N} w_j} \tag{12}$$

where $i = 1, \ldots, 3$, $j = 1, \ldots, N$. $d_{i,j}$ is the distance of the jth component's CG from the origin in the ith axis, w_j is its weight.

The penalty value of this constraint condition is given by the difference in the actual overall CG obtained ($X_{\mathrm{CG}}, Y_{\mathrm{CG}}, Z_{\mathrm{CG}}$) and the stipulated CG location ($X_{\mathrm{CG,s}}, Y_{\mathrm{CG,s}}, Z_{\mathrm{CG,s}}$). The constraints are thus defined as

$$|X_{\mathrm{CG}} - X_{\mathrm{CG,s}}| < \delta_{\mathrm{CG}} \tag{13}$$

$$|Y_{\mathrm{CG}} - Y_{\mathrm{CG,s}}| < \delta_{\mathrm{CG}} \tag{14}$$

$$|Z_{\mathrm{CG}} - Z_{\mathrm{CG,s}}| < \delta_{\mathrm{CG}} \tag{15}$$

In practice, after all the components have been assembled with the wires fitted in, the final CG location may deviate slightly from the stipulated one. Therefore, a slight tolerance of $\delta_{\mathrm{CG}} = 0.001$ m is allowed in these CG constraints. This will avoid the excessive computation in the search to satisfy the more stringent constraints of $X_{\mathrm{CG}} - X_{\mathrm{CG,s}} = 0$, $Y_{\mathrm{CG}} - Y_{\mathrm{CG,s}} = 0$, and $Z_{\mathrm{CG}} - Z_{\mathrm{CG,s}} = 0$. The penalty value of this constraint condition is given by the difference in the actual overall CG obtained by optimization ($X_{\mathrm{CG}}, Y_{\mathrm{CG}}, Z_{\mathrm{CG}}$) and the stipulated CG location ($X_{\mathrm{CG,s}}, Y_{\mathrm{CG,s}}, Z_{CG,s}$)

$$h_4 = (|X_{\mathrm{CG}} - X_{\mathrm{CG,s}}| - \delta_{\mathrm{CG}}) \tag{16}$$

$$h_5 = (|Y_{\mathrm{CG}} - Y_{\mathrm{CG,s}}| - \delta_{\mathrm{CG}}) \tag{17}$$

$$h_6 = (|Z_{\mathrm{CG}} - Z_{\mathrm{CG,s}}| - \delta_{\mathrm{CG}}) \tag{18}$$

3.1.5 *Minimum voltage and current of power source constraint*

Every component from different manufacturers has different power consumption, voltage, and current requirements. Likewise, different electric power packs from different manufacturers have different dimensions, capacity, voltage, and current ratings. It is apparent that the voltage and current ratings of the electric power source must be equal or greater than the components' requirements in order for them to operate normally. In the case where the supply voltage is too high, a simple potential-divider circuit is used to tap the necessary voltage for the component, averting any possible circuit damage to the electrical components. Since the choice of the electric power source is a design variable, it is necessary to enforce the following constraints

$$V_{\mathrm{supply}} \geqslant V_i \tag{19}$$

$$I_{\mathrm{supply}} \geqslant \sum_{i=1}^{N} I_i, \quad i \neq 5 \tag{20}$$

where V_i and I_i are the voltage and current requirements of the ith component, V_{supply} is output voltage and I_{supply} is the output current of the power source.

It is noted that component 5 is the power supply and its power consumption due to internal resistance is treated as negligible here. The penalty value associated with the violated of these two constraints are given by

$$h_7 = \left[\left(\sum_{i=1}^{N} I_i\right) - I_{\mathrm{supply}}\right], \quad i \neq 5 \tag{21}$$

$$h_8 = \sum_{i=1}^{N}(V_i - V_{\mathrm{supply}}) \tag{22}$$

3.1.6 *Minimum lift-to-weight ratio constraint*

Generally, the larger electric motors are more powerful and can generate more thrust. However, they are also heavier and thus contribute to the weight penalty of the vehicle. For a rotary-wing flight vehicle, there is a critical requirement on its propulsion systems. The total thrust generated by the combination of electric motors and propellers must be greater than the total weights of all the components and fuselage, in order for the flight vehicle to be airborne. This essential constraint is defined as

$$\frac{L_{\mathrm{p}}}{W} > 1 + L_{\mathrm{sf}} \tag{23}$$

where L_{sf} is the lift safety factor, L_{p} is the total thrust generated by the propulsion units, W is the total weight of components and estimated fuselage weight. Instead of enforcing this equality constraint, a small tolerance δ can be allowed by the designer

$$\left|(1 + L_{\mathrm{sf}}) - \frac{L_{\mathrm{p}}}{W}\right| < \delta \tag{24}$$

If this constraint is violated, the penalty value is given by

$$h_9 = \left|(1 + L_{\mathrm{sf}}) - \frac{L_{\mathrm{p}}}{W}\right| - \delta \tag{25}$$

3.1.7 *Minimum flight time constraint*

It seems that all the necessary constraints have been considered after the components are positioned as close as possible without any overlapping

with the overall CG, minimum thrust and electrical requirements fulfilled.

However, it would serve no purpose if the UAV/MAV can only fly for a very short period to accomplish any meaningful mission. Therefore, there is a need to ensure that the flight vehicle is able to perform flight for a minimum stipulated time. The flight time, T_{flight}, can be calculated as follows

$$T_{\text{flight}} = \frac{\text{battery capacity}}{\text{total current usage}} \tag{26}$$

This constraint is given as

$$T_{\text{flight}} \geqslant T_{\text{flight, s}} \tag{27}$$

where T_{flight} is the actual flight time obtained and $T_{\text{flight,s}}$ is the desired minimum flight time. If this constraint is violated, the penalty value is given by

$$h_{10} = T_{\text{flight,s}} - T_{\text{flight}} \tag{28}$$

3.1.8 Balanced yaw moments

If the four propulsion sets are identical, and each set generate 1/4 of the total lift required during hovering, the reaction torques will be of same magnitude and cancel each other. However, if the two pairs are not identical, it is not just sufficient for them to produce a total lift that is equal to weight. At this instance, the reaction torques generated by the two different pairs must also cancel one another, so that there will be no net yaw moment that will cause the vehicle to spin around. This consideration is essential since there is no tail rotor in the quadrotor configuration that is responsible for the yaw torque cancellation. Whether or not it is possible to have two pairs of different propulsion sets that can fulfil these conditions depending on the hardware availability, and this possibility should not be ruled out. The constraint is defined as

$$\sum \text{Moment}_{\text{yaw}} = 0 \tag{29}$$

Instead of enforcing this equality constraint, a small tolerance δ can be allowed by the designer

$$\left| \sum \text{Moment}_{\text{yaw}} \right| < \delta \tag{30}$$

If this constraint is violated, the penalty value is given by

$$h_{11} = \left| \sum \text{Moment}_{\text{yaw}} \right| - \delta \tag{31}$$

As mentioned earlier, because the two pairs of propulsion sets are arranged in symmetry, there is no resultant pitch and roll moments, when each motor of the same pair produces the same amount of thrust during hovering. Hence, there is no need for two other constraints to enforce zero resultant pitch and roll moments.

4 GENETIC ALGORITHMS

GA consist of search procedures that mimic the natural biological evolution (Holland [**12**]), with many improvements contributed subsequently (De Jong [**13**]; Goldberg [**14**]). GA operate on a population of potential solutions, applying the 'survival of the fittest' principle to produce better approximations to a solution. Solutions in GA are represented as chromosomes with the most common representation using a single-level, binary string.

Studies done (Wright [**15**]) have shown that real-valued representation in GA offers numerous advantages such as better efficiency and lesser computer memory usage. Thus, real-valued representation is used in this study. Each of the design parameters is represented by a real number and the chromosome is an array of these design parameters, rather than a binary string. The parameters have their corresponding upper and lower bounds of allowable values. One obvious advantage is the reduction of overheads incurred in the conversion of the binary number into the real number. In the current study, the chromosome is a string of mixed combination of integer and float numbers. The objective function in the GA optimization problem is defined as

$$\begin{aligned}\text{minimize } f = \sum_{i=1}^{11} \sigma_i h_i + \sigma_{12}(&X_{\text{fuselage}} \times Y_{\text{fuselage}} \times Z_{\text{fuselage}} \\ &+ X_{\text{propeller}} \times Y_{\text{propeller}} \times Z_{\text{propeller}})\end{aligned} \tag{32}$$

where $h_{i=1 \text{ to } 11}$ are the penalty functions defined earlier for the constraints and $\sigma_{i=1 \text{ to } 12}$ are the weighing factors.

Modular GA written in C programming language consists of the following sub-modules: selection, crossover, and mutation. In GA, the selection operator determines the number of times a parent chromosome is chosen for reproduction. There are many variations concerning how potential parents are selected to produce new offspring. The method chosen here is one of the most commonly used methods, *roulette wheel selection*, where chromosomes with high fitness values have a higher probability in participating in reproduction of new chromosomes. This mimics the 'survival of the fittest' phenomenon that one can observe in nature. The probability of an individual chromosome selected for reproduction is given by

$$P_i = \frac{f_i}{\sum_{i=1}^{N_{\text{chrom}}} f_i} \tag{33}$$

where f_i is the fitness of ith individual chromosome, and N_{chrom} is total number of chromosomes in the population.

Crossover is the main genetic search operator in GA. It recombines the genetic material in two parent chromosomes to create two children chromosomes. The new individuals possess some parts of genetic material from both the parent chromosomes. It is a powerful process that extends the search in many directions. The crossover operation is controlled by a crossover rate, denoted by P_c, which is usually set at 80 to 100 per cent. Intermediate crossover is adopted in this study as studies done (Muhlenbein and Schlierkamp-Voosen [**16**]) have revealed promising results.

Mutation serves as a secondary search operator for exploring new search region by altering the allele of a gene randomly. For real-valued chromosomes, mutation is achieved by perturbing the gene values within the allowed range. This operation helps to ensure that the search process will not be trapped in local minima. The probability (or mutation rate), denoted by P_m, of the mutation process is usually set low, typically around 5 to 20 per cent. It is not advisable to set P_m too high, as the optimization process will tend towards a haphazard and random search.

There is no standard criterion for terminating the search process. It really depends on the amount of time the designer is willing to wait to obtain the near-optimal or exact optimal solutions. In this study, the termination criterion is set to be the point when the improvement in objective value is less than 0.5 per cent for consecutive 100 000 generation. The optimization design flowchart using GA is shown in Fig. 6.

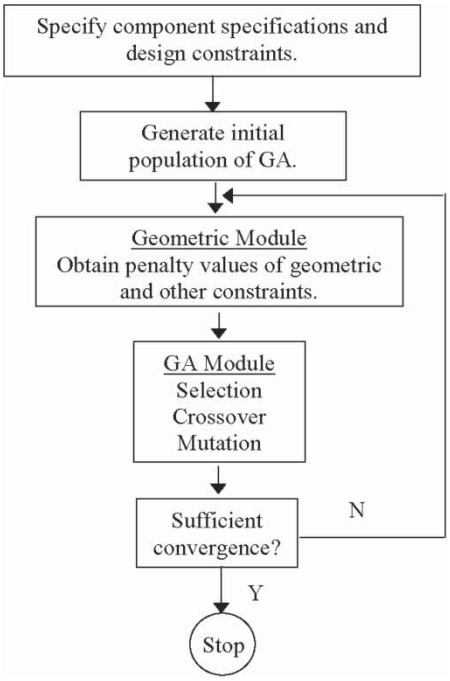


Fig. 6 Flow chart of design optimization using GA

5 RESULTS

5.1 Case study number 1

To investigate the feasibility of the proposed design strategy, the methodology is applied to the design of the quadrotor rotary-wing UAV/MAV using 11 small-scale commercial-off-the-shelf components: four electric propulsion sets, radio receiver, pitch rate sensor, roll rate sensor, yaw rate sensor, electric power-pack, video transmitter, and video camera. (Refer to Tables 1–3 for the available components' specifications).

There are 45 design variables consisting of 20 integer variables and 25 float number variables, whose lower and upper bounds are shown in Table 4. In this case study, the desired overall CG location has been arbitrarily chosen to be at the origin, i. e. ($X_{CG,s}$, $Y_{CG,s}$, $Z_{CG,s}$) is (0, 0, 0). The minimum required clearance between the propellers, $\delta_{prop,s}$ is 0.01 m, minimum desired flight

Table 1 Table of specifications of available propulsion sets

Component	L (m)	B (m)	H (m)	M (kg)	Propeller diameter (m)	Propeller height (m)	Minimum voltage (V)	Maximum voltage (V)	Thrust-to-current ratio (kg/A)	Torque-to-current ratio (Nm/A)	Voltage-to-current ratio (V/A)
Propulsion set number 1	0.012	0.010	0.015	0.006	0.065	0.008	4.0	8.0	0.03	0.01	10.18
Propulsion set number 2	0.012	0.010	0.015	0.0076	0.200	0.015	4.0	8.0	0.0595	0.021	10.18
Propulsion set number 3	0.012	0.012	0.018	0.0126	0.200	0.015	4.0	10.0	0.0837	0.020	9.22
Propulsion set number 4	0.012	0.012	0.018	0.0132	0.230	0.015	4.0	10.0	0.0797	0.026	9.00
Propulsion set number 5	0.012	0.012	0.018	0.0136	0.260	0.015	4.0	10.0	0.0904	0.034	11.96
Propulsion set number 6	0.017	0.017	0.017	0.0206	0.260	0.015	4.0	10.0	0.1056	0.032	12.52
Propulsion set number 7	0.015	0.015	0.024	0.0249	0.280	0.015	4.0	10.0	0.1444	0.036	13.16

Table 2 Table of specifications of available electric power sources

Component	L (m)	B (m)	H (m)	M (kg)	Voltage output (V)	Current output (A)	Capacity (Ah)
Electric power source number 1	0.045	0.015	0.051	0.051	9.0	2.0	0.770
Electric power source number 2	0.015	0.015	0.153	0.051	9.0	2.0	0.770
Electric power source number 3	0.045	0.015	0.034	0.036	9.0	1.0	0.425
Electric power source number 4	0.015	0.015	0.102	0.036	9.0	1.0	0.425
Electric power source number 5	0.070	0.028	0.014	0.055	9.6	2.0	0.110
Electric power source number 6	0.045	0.030	0.051	0.102	9.0	4.0	1.540
Electric power source number 7	0.030	0.015	0.102	0.102	9.0	4.0	1.540
Electric power source number 8	0.070	0.028	0.028	0.110	9.6	4.0	0.220
Electric power source number 9	0.036	0.021	0.022	0.028	11	3.6	0.350

Table 3 Table of technical specifications of other components

Component	L (m)	B (m)	H (m)	M (kg)	Voltage (V)	Current (A)
Rate sensor number 1	0.025	0.025	0.013	0.009	3.0	0.014
Rate sensor number 2	0.027	0.026	0.012	0.005	4.8	0.010
Radio receiver number 1	0.027	0.015	0.006	0.003	5.0	0.010
Radio receiver number 2	0.036	0.017	0.012	0.006	5.0	0.010
Video camera	0.026	0.026	0.015	0.012	9.0	0.100
Video transmitter	0.030	0.018	0.010	0.007	9.0	0.300

time, $T_{\text{flight,s}}$ is 300 s and the lift safety factor, L_{sf} is 0.05. These user-input design constraints are summarized in Table 5. The estimated mass of fuselage together with some miscellaneous components is 0.014 kg. The GA parameters are population size of 100, crossover probability, P_{c}, of 0.90, and mutation probability, P_{m}, of 0.20. The design optimizations are performed using a Pentium IV 2.4 GHz PC, for 400 000 generations (iterations).

At the first generation, only three out of the eight constraints are violated (Table 6). From Fig. 7, it can be seen that the components are well spread out, and thus do not overlap with one another. However, the CG location obtained is not at the desired origin. The front/back propeller is 23 cm while the left/right propeller is 26 cm in diameter. The total lift generated is also lesser than the total mass, and thus the solution at this instance is still not a feasible one. This comes as no surprise as this is only the first generation and better results can be expected.

After 523 generations, the design parameters have evolved to produce a feasible solution with no constraint being violated. Both pairs of propulsion sets are number 4, whose propeller diameter is 23 cm. The overall dimensions of the current vehicle are tabulated in Table 7. From Fig. 8, it can be seen that the propellers are placed very closely but the components are still not organized in a very compact manner yet.

At the 379 928th generation, the design parameters have further evolved, with the most significant change in the propulsion sets from number 4 to number 3 which have a propeller diameter of 20 cm (Fig. 9), and the corresponding overall dimensions are summarized in Table 8.

Finally, the best optimal results are obtained in the 380 710th generation. Figure 10 shows the final optimal layout and the optimal dimensions are tabulated in Table 9. The values of the corresponding optimal design variables are given in Table 4. The final propulsion sets chosen are still number 3, with each set contributing 1/4 of the total required lift. It is interesting to note that set number 1, which has the smallest propeller of the available choices, is not selected. This is because four number 1 propulsion sets are not able to produce enough lift to support the total weight of the vehicle. It seems that though one pair of set number 3 and one pair of set number 1 are able to generate sufficient lift force, it is, unfortunately, that this combination is not able to cancel each other's reaction torques.

The electric power source selected is the heaviest battery number 7 instead of the lightest battery number 9. The quadrotor has four heavy identical propulsion sets, and the total mass is quite high. Thus, it needs to consume a very high current in order to generate the necessary lift. Battery number 9 is unable to meet the current demand and is thus not chosen.

There is another very interesting observation in the results obtained. GA has managed to select the lighter radio receiver number 1 instead of heavier radio receiver number 2. However, the pitch and yaw gyros selected are the heavier gyro number 1 while the roll gyro is the lighter gyro number 2. At first, this may appear to be a strange outcome, but a closer examination would help explain this mystery. The objective of the optimization program is to minimize the overall dimensions, without violating any of the eight design constraints. By replacing the pitch and yaw gyros with a lighter gyro number 2, it would not be able to produce a more compact vehicle, and so there is no 'motivation' for GA to change those particular design variables,

Table 4 Table of design variables, corresponding bounds, and optimal results

	Variable name	Lower bound	Upper bound	Best optimal results
1	Front/back propulsion choice	1	7	3
2	Left/right propulsion choice	1	7	3
3	Front motor orientation	1	2	2
4	Back motor orientation	1	2	1
5	Left motor orientation	1	2	1
6	Right motor orientation	1	2	1
7	Front propulsion x position (m)	0.001	0.500	0.182
8	Left propulsion y position (m)	0.001	0.500	0.105
9	Front/back propulsion percentage usage of maximum allowable current	0.010	0.990	0.672
10	Left/right propulsion percentage usage of maximum allowable current	0.010	0.990	0.689
11	Radio receiver choice	1	2	1
12	Radio receiver mounting plane	1	3	1
13	Radio receiver orientation	1	2	1
14	Radio receiver x position (m)	−0.500	0.500	−0.047
15	Radio receiver y position (m)	−0.500	0.500	0.050
16	Radio receiver z position (m)	−0.500	0.500	0.013
17	Pitch rate sensor choice	1	2	1
18	Pitch rate sensor mounting plane	1	2	1
19	Pitch rate sensor x position (m)	−0.500	0.500	−0.019
20	Pitch rate sensor y position (m)	−0.500	0.500	0.048
21	Pitch rate sensor z position (m)	−0.500	0.500	−0.002
22	Roll rate sensor choice	1	2	2
23	Roll rate sensor mounting plane	1	2	1
24	Roll rate sensor x position (m)	−0.500	0.500	−0.153
25	Roll rate sensor y position (m)	−0.500	0.500	0.069
26	Roll rate sensor z position (m)	−0.500	0.500	−0.003
27	Yaw rate sensor choice	1	2	1
28	Yaw rate sensor orientation	1	2	2
29	Yaw rate sensor x position (m)	−0.500	0.500	0.000
30	Yaw rate sensor y position (m)	−0.500	0.500	0.085
31	Yaw rate sensor z position (m)	−0.500	0.500	0.003
32	Electric power source choice	1	9	7
33	Electric power source mounting plane	1	3	3
34	Electric power source orientation	1	2	2
35	Electric power source x position (m)	−0.500	0.500	−0.001
36	Electric power source y position (m)	−0.500	0.500	−0.023
37	Electric power source z position (m)	−0.500	0.500	−0.003
38	Video transmitter mounting plane	1	3	1
39	Video transmitter orientation	1	2	2
40	Video transmitter x position (m)	−0.500	0.500	−0.038
41	Video transmitter y position (m)	−0.500	0.500	−0.073
42	Video transmitter z position (m)	−0.500	0.500	−0.004
43	Video camera x position (m)	0.000	0.500	0.121
44	Video camera y position (m)	−0.500	0.500	0.080
45	Video camera z position (m)	−0.500	0.500	0.004

Table 5 Table of design constraints requiring designer inputs

Design constraints	Value
Stipulated minimum clearance between propellers, $\delta_{\text{prop,s}}$	0.01 m
Stipulated overall CG x location, $X_{\text{CG,s}}$	0.00 m
Stipulated overall CG y location, $Y_{\text{CG,s}}$	0.00 m
Stipulated overall CG z location, $Z_{\text{CG,s}}$	0.00 m
Lift safety factor, L_{sf}	0.05
Desired minimum flight, $T_{\text{flight,s}}$	300 s

Table 6 Status of optimization constraints at first generation

Constraints	Status
Overlapping regions	Fulfilled
Maximum Z boundary	Violated
Overall CG location	Violated
Inter-propeller distance	Fulfilled
Balanced yaw moment	Fulfilled
Minimum voltage and current of power source	Fulfilled
Lift-to-weight ratio	Violated
Minimum flight time	Fulfilled

since it has already accomplished its goal. In fact, if those variables were changed to the lighter gyro number 2 and their location variables are not changed appropriately, the CG location constraint would have been violated. Another possible reason is that the extra mass incurred by having two heavier gyros is only

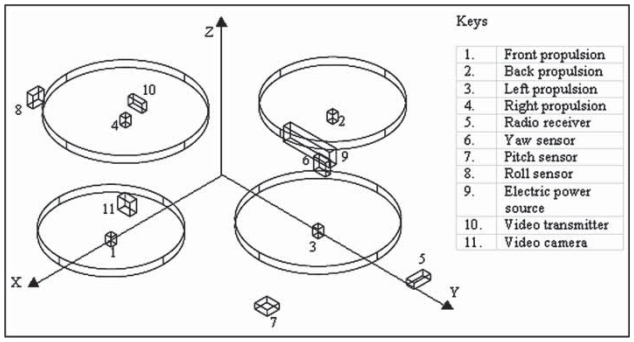


Fig. 7 Layout obtained by optimization at first generation

Table 7 Results of overall dimensions at 523rd generation

Dimension (m)	Value
Overall fuselage x	0.3320
Overall fuselage y	0.3700
Overall fuselage z	0.1160
Overall propellers' x	0.5500
Overall propellers' y	0.5880
Overall propellers' z	0.0150

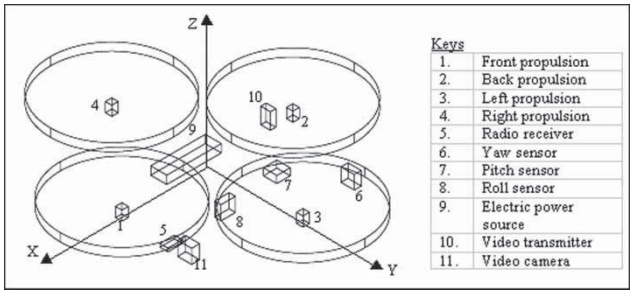


Fig. 8 Layout obtained by optimization at 523rd generation

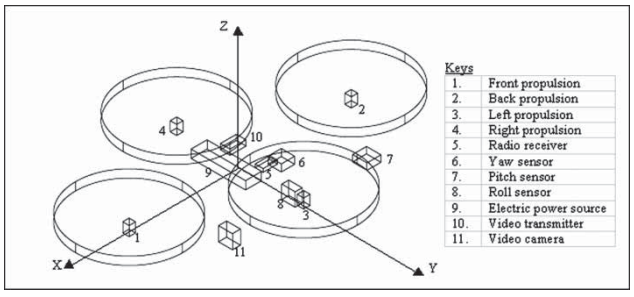


Fig. 9 Layout obtained by optimization at 379 928th generation

0.008 kg, which is not very significant considering that the quadrotor is able to provide ample lift force.

The total weight of the vehicle is 2.178 N, with all the other constraints fulfilled as well. This combination of components is able to achieve a theoretical flight time of 1785 s.

Table 8 Results of overall dimensions at 379 928th generation

Dimension (m)	Value
Overall fuselage x	0.4280
Overall fuselage y	0.2500
Overall fuselage z	0.0325
Overall propellers' x	0.6160
Overall propellers' y	0.4380
Overall propellers' z	0.0150

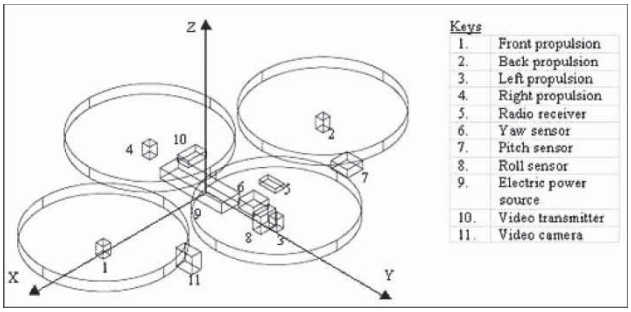


Fig. 10 Final layout obtained by optimization at 380 710th generation

Table 9 Final optimal overall dimensions at 380 710th generation

Dimension (m)	Value
Overall fuselage x	0.3760
Overall fuselage y	0.2220
Overall fuselage z	0.0285
Overall propellers' x	0.5640
Overall propellers' y	0.4100
Overall propellers' z	0.0150

5.2 Case study number 2 – quality of optimization convergence

It is inevitable that the final solutions from different runs using different initial GA populations are not identical, due to the large number of design variables and the stochastic nature of GA. The optimization problem in case study number 1 is repeated ten times and the variations in the optimal results from the multiple runs reveal that the difference between the worst and best results is only 11 per cent, acceptable for engineering practicality (Table 10).

5.3 Case study number 3 – trends of increasing payload

Using the same GA parameters and design constraints mentioned in case study number 1, a follow-up

Table 10 Variations in final optimal results obtained for ten runs

Run	Fuselage x dimension (m)	Fuselage y dimension (m)	Fuselage z dimension (m)	Overall propeller's x dimension (m)	Overall propeller's y dimension (m)	Overall propeller's z dimension (m)	Optimal value ($\times 10^{-3}$ m^3)
1	0.3760	0.2220	0.0285	0.5640	0.4100	0.0150	5.85
2	0.3760	0.2240	0.0355	0.5660	0.4120	0.0150	6.49
3	0.3760	0.2220	0.0295	0.5640	0.4100	0.0150	5.93
4	0.3760	0.2220	0.0285	0.5640	0.4100	0.0150	5.85
5	0.3760	0.2220	0.0285	0.5640	0.4100	0.0150	5.85
6	0.3760	0.2220	0.0300	0.5640	0.4100	0.0150	5.97
7	0.3760	0.2220	0.0295	0.5640	0.4100	0.0150	5.93
8	0.3760	0.2220	0.0285	0.5640	0.4100	0.0150	5.85
9	0.3760	0.2220	0.0300	0.5640	0.4100	0.0150	5.97
10	0.3760	0.2220	0.0285	0.5640	0.4100	0.0150	5.85

study is carried out to investigate the trends in the overall vehicle dimensions and flight time versus increasing payload. The only difference from the earlier case study is that a dummy payload of dimensions 0.020 m × 0.020 m × 0.020 m with mass 0.010 kg is added in this study. The dimensions of this dummy payload remain fixed but its mass will increase incrementally by 0.010 kg to determine the trends.

Figure 11 shows that the overall vehicle volume generally increases with an increase in mass of payload, which is consistent with real-life experiences. In Fig. 12, the graph reveals that there is a general trend of shorter flight time with increasing payload, as expected. There is an abrupt drop in flight time when the payload reaches 0.070 kg. This is because the lighter electric power source number 9 is now chosen in preference to the heavier electric power source number 7. However, the capacity of electric power source number 9 is significantly lower than number 7, resulting in a sudden drop in flight time, but still able to satisfy the minimum flight time requirement of 300 s.

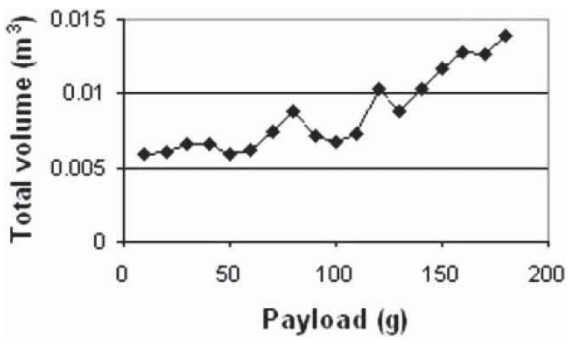


Fig. 11 Graph of total volume versus payload

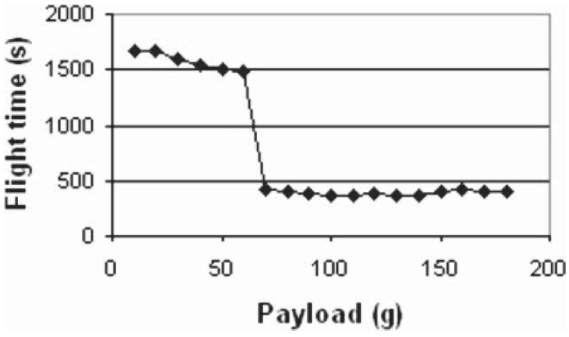


Fig. 12 Graph of flight time versus payload

6 CONCLUSIONS

In the current paper, a design methodology has been introduced to automate the component selection, configuration layout and geometric sizing of a small-scale quadrotor UAV/MAV, and has been implemented with success.

The design optimization methodology allows the designer to use a standard personal computer to solve the complex design problem, and produce a satisfactory preliminary configuration much faster than the usual trial-and-error approach.

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APPENDIX

Notation

B	breadth of component (m)
BH	surface of the component comprising of its breadth and height
BL	surface of the component comprising of its breadth and length
CG	centre of gravity of the UAV/MAV (m)
$CG_{component}$	centre of gravity of individual component (m)
f_i	fitness value of ith individual chromosome
H	height of component (m)
HL	surface of the component comprising of its height and length
I_i	current requirement of ith component (A)
I_{supply}	current output of electric power supply (A)
K_{thrust}	thrust versus current constant (N/A)
K_{torque}	torque versus current constant (Nm/A)
K_v	voltage versus current constant (V/A)
L	length of component (m)
L_p	lift generated by propellers (N)
L_{sf}	lift safety factor
M	mass of component (kg)
N	total number of components in UAV/MAV
N_{chrom}	total number of chromosomes in the population
P_c	probability of crossover
P_i	probability of ith chromosome selected for reproduction
P_m	probability of mutation
T_{flight}	theoretical flight time of UAV/MAV (s)
$T_{flight,s}$	stipulated minimum flight time of UAV/MAV (s)
V_i	voltage requirement of ith component (V)
V_{supply}	voltage output of electric power supply (V)
$Vol_{overlap,i}$	volume of ith component overlapped with others (m^3)
$Vol_{protrude,i}$	volume of ith component protruding the rotor/propeller plane (m^3)
W	total weight of flight vehicle (N)
x	position of the component's centre of gravity with respect to the X-axis (m)
X_{CG}	position of the overall CG obtained by optimization with respect to the X-axis (m)
$X_{CG,s}$	position of the desired overall CG with respect to the X-axis (m)
$X_{fuselage}$	overall x dimension of the UAV's fuselage (m)
$X_{propeller}$	overall x dimension of the UAV's propellers (m)
y	position of the component's centre of gravity with respect to the Y-axis (m)
Y_{CG}	position of the overall CG obtained by optimization with respect to the Y-axis (m)
$Y_{CG,s}$	position of the desired overall CG with respect to the Y-axis (m)
$Y_{fuselage}$	overall y dimension of the UAV/MAV's fuselage (m)
$Y_{propeller}$	overall y dimension of the UAV/MAV's propellers (m)
z	position of the component's centre of gravity with respect to the Z-axis (m)

Z_{CG}	position of the overall CG obtained by optimization with respect to the Z-axis (m)
$Z_{CG,s}$	position of the desired overall CG with respect to the Z-axis (m)
$Z_{fuselage}$	overall z dimension of the UAV/MAV's fuselage (m)
$Z_{propeller}$	overall z dimension of the UAV/MAV's propellers (m)
Z_{max}	z plane of rotor/propeller (m)
δ_{CG}	tolerance in deviation of overall centre of gravity obtained (m)
δ_{prop}	distance between adjacent propellers (m)
$\delta_{prop,s}$	stipulated minimum distance between adjacent propellers (m)